Seneca Polytechnic | Work-Integrated Learning (WIL)

INDIVIDUALIZED LEARNING PLAN (ILP)

- **Student Name:** Chao Chen
- **Student Number:** *135082196*
- **Program of Study:** Applied Artificial Intelligence & Machine Learning / Software Design
- **Host Organization:** Seneca HELIX (Entrepreneurship Work-Integrated Learning – EWIL Strand)
- **Project / Venture Title:** Chronicles of the Loom (AI-Powered Emergent Narrative Simulation & TRPG System)
- **Placement Role:** Lead Technical Designer & AI Systems Developer

GOAL #1: Technical & Software System Architecture (Program Learning Outcome Goal)

- **Learning Goal (I want to learn...):**
  I want to master the design and implementation of a modular, decoupled 4-layer system architecture for real-time emergent narratives, focusing specifically on integrating low-latency AI decision loops with transparent causal math algorithms. I want to ensure that complex agent memory states and social relationship dynamics can be executed smoothly without causing performance bottlenecks in the UI/data rendering layer.
- **Strategy (I will learn by...):**
  I will learn by applying clean code design patterns (such as the Observer pattern and Entity-Component decoupled structures) in Python/C#, conducting benchmark tests on real-time event loops, and reviewing algorithmic pipelines with technical mentors at Seneca HELIX. I will also study advanced gameplay architectures from technical post-mortems of systemic games like *Caves of Qud* and *RimWorld*.
- **What Success May Look Like (Accomplishing my goal may look like...):**
  Accomplishing this goal will be demonstrated by delivering a functional, self-contained interactive prototype featuring 10 active NPCs who execute social interactions, grudge tracking, and item-sharing driven by fully exposed mathematical logic. Success means achieving zero state desynchronization and maintaining sub-second latency during NPC decision calculations.
- **Target Date:**
  October 23, 2026 (Mid-Term Milestone)
- **Post Co-op/Work Term Evidence of Success (For post-term reflection / Mid-term site visit):**

I successfully architected and stress-tested the decoupled event pipeline, reducing calculation latency under 800ms. The prototype remained completely crash-free during internal multi-agent simulations, and the technical documentation was reviewed and validated by our technical advisor.

GOAL #2: Customer Discovery & Qualitative Market Validation (EWIL Milestone Goal)

- **Learning Goal (I want to learn...):**
  I want to develop professional user research and customer discovery skills by conducting structured, non-leading qualitative interviews with target gamer segments (Colony Sim fans, Hardcore Systems Thinkers, and Tabletop RPG players). I aim to learn how to translate raw player feedback and frustration regarding "black-box mechanics" into actionable software feature requirements.
- **Strategy (I will learn by...):**
  I will learn by executing the 10-question customer validation framework developed in HELIX EWIL Workbook 5. I will proactively reach out to niche gaming communities across Discord and Reddit, conduct at least 10 to 15 one-on-one recorded interviews, and systematically log their behavioral habits, price sensitivities, and UI expectations into a structured validation matrix.
- **What Success May Look Like (Accomplishing my goal may look like...):**
  Accomplishing this goal means producing a comprehensive Customer Validation Report that consolidates feedback across three distinct user personas. Success will also include securing at least 5 dedicated external playtesters who commit to participating in our upcoming pre-alpha build tests via screen share.
- **Target Date:**
  November 6, 2026
- **Post Co-op/Work Term Evidence of Success (For post-term reflection / Mid-term site visit):**
  Completed 15 qualitative customer validation interviews, which directly informed our UI/UX redesign from overwhelming raw logs to intuitive causal chains. Documented findings in the customer matrix and successfully onboarded early playtesters into a dedicated feedback channel.

GOAL #3: Professional Networking & Event Strategy Formulation (Venture Growth Goal)

- **Learning Goal (I want to learn...):**

I want to learn how to design and execute a comprehensive go-to-market events and industry exhibition strategy (Events & Exhibits Support Plan) to showcase an early-stage venture to investors, industry peers, and potential technical collaborators.

- **Strategy (I will learn by...):**
  I will learn by analyzing regional gaming conventions and technology summits in Canada (such as Toronto Game Expo, Seneca HELIX Demo Day, Breakout Con, and XP Game Summit). I will evaluate each venue's cost-to-reach metrics, attendee profiles, booth packages, and pitch formats, and incorporate mentorship feedback from HELIX venture advisors to refine our commercial pitch deck.
- **What Success May Look Like (Accomplishing my goal may look like...):**
  Accomplishing this goal will be shown by completing a fully costed, dated Event & Exhibit Road-Map for 2026–2027 with direct registration and budget breakdowns. Additionally, delivering an executive 5-minute milestone presentation deck suitable for the HELIX Demo Day & Founders Mixer will demonstrate milestone completion.
- **Target Date:**
  November 20, 2026
- **Post Co-op/Work Term Evidence of Success (For post-term reflection / Mid-term site visit):**
  Finalized the 2026–2027 exhibition plan covering 6 key regional and national industry events, with clear budget allocations. Rehearsed and delivered our venture presentation at the Seneca HELIX incubator review, receiving positive validation on our market entry approach.

GOAL #4: Agile Self-Management & Professional Communication (Professional Development Goal)

- **Learning Goal (I want to learn...):**
  I want to strengthen my self-directed project management, time-blocking, and upward professional communication skills. Specifically, I want to learn how to balance intensive sprint programming with non-technical founder responsibilities, while keeping academic and venture stakeholders consistently informed.
- **Strategy (I will learn by...):**
  I will learn by adopting a structured two-week sprint methodology using digital project tracking tools (such as Jira/Kanban boards). I will schedule recurring weekly syncs with my WIL coordinator/mentors, proactively report milestones, highlight blockers early, and strictly maintain structured reflective documentation following the L.E.A.R.N. model.
- **What Success May Look Like (Accomplishing my goal may look like...):**
  Accomplishing this goal will be reflected in 100% on-time submission of all work term assignments (ILP, Discussion Boards, and monthly LEARN reflections), maintaining updated sprint backlogs without unresolved blockers lingering for more than 48 hours, and

receiving positive performance feedback during the Mid-Semester Check-in.

- **Target Date:**
  December 4, 2026 (End of Term Wrap-up)
- **Post Co-op/Work Term Evidence of Success (For post-term reflection / Mid-term site visit):**
  Maintained consistent weekly sprint updates and delivered all course assignments ahead of deadlines. Successfully utilized the L.E.A.R.N. model to analyze and correct workflow bottlenecks, receiving an 'Exceeds Expectations' rating in the Mid-Semester review for self-direction and stakeholder transparency.